Introduction to Machine Learning ? Lecture 1

Welcome to CS 4780: Introduction to Machine Learning. In this course we will study how computers learn from data to make predictions and decisions without being explicitly programmed.

Machine learning is broadly divided into three paradigms: supervised learning, unsupervised learning, and reinforcement learning. Today's lecture focuses on supervised learning.

In supervised learning, we are given a training set of labelled examples {(x_1, y_1), ..., (x_n, y_n)}. The goal is to learn a function f such that f(x) ? y for new, unseen inputs x.

Common supervised learning tasks include classification (predicting a discrete label) and regression (predicting a continuous value). For example, classifying whether an email is spam or not spam is a classification problem.

The bias-variance trade-off is a fundamental concept in machine learning. Bias refers to error introduced by approximating a complex real-world problem with a simplified model. Variance refers to sensitivity of the model to fluctuations in the training set.

High bias leads to underfitting ? the model is too simple to capture the underlying pattern. High variance leads to overfitting ? the model memorises training data but fails to generalise.

Linear regression is one of the simplest supervised learning algorithms. Given features x ? R^d and targets y ? R, it fits a linear function f(x) = w^T x + b by minimising the mean squared error (MSE) loss.

The solution can be found in closed form via the normal equations: w = (X^T X)^{-1} X^T y, or iteratively using gradient descent.

Gradient descent is an iterative optimisation algorithm. Starting from an initial parameter vector w_0, we repeatedly update: w_{t+1} = w_t - ? ?L(w_t), where ? is the learning rate and ?L is the gradient of the loss.

Stochastic gradient descent (SGD) approximates the full gradient using a single random sample or a mini-batch, which is computationally efficient for large datasets.

Decision trees partition the feature space into rectangular regions. At each internal node the tree selects a feature and threshold that best splits the data, typically by minimising Gini impurity or maximising information gain.

Random forests are ensembles of decision trees trained on bootstrap samples of the data. By averaging predictions, they reduce variance while keeping bias low.

Support vector machines (SVMs) find the maximum-margin hyperplane that separates two classes. The margin is the distance between the hyperplane and the nearest training examples, called support vectors.

For non-linearly separable data, the kernel trick allows SVMs to operate in a high-dimensional feature space implicitly, without computing the explicit mapping.

Model evaluation is critical. We must not evaluate a model on the data it was trained on, as this gives an overly optimistic estimate. Instead, we hold out a test set.

Cross-validation ? particularly k-fold cross-validation ? is a robust evaluation strategy. The training data is split into k folds; the model is trained on k-1 folds and evaluated on the remaining fold, cycling through all folds.

Regularisation is a technique to reduce overfitting by adding a penalty term to the loss function. L2 regularisation (Ridge) adds ?||w||_2^2, shrinking weights towards zero. L1 regularisation (Lasso) adds ?||w||_1, which can drive some weights to exactly zero, producing sparse models.

The regularisation strength ? is a hyperparameter that must be tuned using a validation set.

Summary: Today we covered the supervised learning paradigm, the bias-variance trade-off, linear regression, gradient descent, decision trees, random forests, SVMs, model evaluation, and regularisation. Next lecture we will explore neural networks and deep learning.